# CivicShield — Technical Audit & Specification

**AI-Enabled Verified Official Identity & Safe Mobile Urban Sensing Platform • SIH 2026 PS 26124**

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| **Target Event:** Smart India Hackathon 2026 | **Problem Statement:** PS 26124 |
| **Primary Backend:** Python FastAPI (SQLite / SQLAlchemy) | **Primary Frontend:** Next.js 14 (React, TypeScript) |
| **AI Models:** IsolationForest, RandomForest, DBSCAN, Gemini | **GIS Engine:** Leaflet / CartoDB Voyager |

## 1. Codebase Inspection & Audit Findings

An empirical audit of the civicAI repository confirms a fully functional web architecture. The codebase is cleanly split into a FastAPI backend service and a Next.js 14 frontend application.

- **Backend Architecture:** FastAPI framework with 10 API routers, SQLAlchemy ORM layer, local SQLite database (civicai.db), 7 ML modules, and WebSocket live telemetry manager.
- **Frontend Architecture:** Next.js 14 App Router featuring 30 compiled page routes, TypeScript type safety, Leaflet map rendering, and HTML5 QR camera scanning.

## 2. Key Features Implementation Breakdown

### A. Official Identity Verification (/verify)

Uses browser camera scanning (html5-qrcode) or manual ID entry to query /api/officers/{officer_id}/verify. Returns official designation, trust score out of 100, verification status badge (ACTIVE, SUSPENDED), and anti-extortion warnings.

### B. Citizen Rating & Anti-Abuse Engine (/ratings)

Form submits weighted ratings across 4 metrics. Evaluates ratings through an **IsolationForest ML model** (rating_anomaly.py) to flag burst manipulation from duplicate IPs/devices.

### C. Government Office & Service Locator (/offices)

Queries /api/offices/nearby using Python-calculated **Haversine geographical distance formulas**. Filters offices by category (Police, Hospital, EB, Municipality, SRO) and displays distance-sorted pins.

### D. Smart Bus & Mobile Urban Sensing Platform (/sih-sensing)

Implements the SIH PS 26124 Mobile Urban Sensing Engine featuring a 4-camera onboard feed switcher (Front, Rear, Side, Cabin), TensorRT YOLOv8 road defect overlays (Pothole 94.8%), ANPR plate extraction, pedestrian safety alerts, and central Leaflet GIS hazard map auto-pinning with automated GCC municipal repair tickets.

### E. Multimodal Travel Planner (/mobility, /trip-manager)

Provides route fare and duration solvers comparing bus, metro, cab, and auto modes. Integrates **Google Gemini 1.5 Flash API** for conversational natural language trip itineraries.

## 3. AI / ML Capabilities Implementation Matrix

| Feature | Algorithm / Model | Input → Output | Status |
|---|---|---|---|
| Rating Anti-Abuse | IsolationForest | 8 Interaction Features → Anomaly Score (-1/1) | **IMPLEMENTED** |
| Passenger Demand | RandomForestRegressor | Route, Hour, Day, Weather → Count | **IMPLEMENTED** |
| O-D Patterns | DBSCAN Clustering | Origin/Dest Coordinates → Cluster Label | **IMPLEMENTED** |
| Office Queue Time | Linear Regression | Visitors, Hour, Staff → Wait Mins | **IMPLEMENTED** |
| Trip Assistant | Google Gemini 1.5 Flash | Natural Language Query → Formatted Itinerary | **IMPLEMENTED** |
| Mobile Fleet Sensing | YOLOv8s + LPRNet Sim | 1080p Video → BBox, Class, Conf, GPS | **PROTOTYPE** |

## 4. Frontend & Backend Technology Stack

| Layer | Technologies Present | Version / Details |
|---|---|---|
| Frontend Framework | Next.js (App Router), React, TypeScript | Next.js 14.2.35, React 18.3 |
| Styling & Icons | Tailwind CSS, Lucide React | Tailwind 3.4, Lucide 0.453 |
| Geospatial / GIS | Leaflet, React-Leaflet, CartoDB Tiles | Leaflet 1.9.4 |
| Backend Framework | Python FastAPI, Uvicorn ASGI | FastAPI 0.110+ |
| ORM & Database | SQLAlchemy 2.0, SQLite Engine | civicai.db local database |
| WebSockets | FastAPI ConnectionManager | ws://127.0.0.1:8000/ws/buses |

## 5. Database, Integrations & Deployment

• **Database Layer:** Relational SQLite database file (civicai.db) managed via SQLAlchemy ORM containing 8 tables (User, Official, Rating, GovtOffice, BusRoute, BusStop, LiveVehicle, Grievance).

• **Live Integrations:** Google Gemini 1.5 Flash API (Live), Leaflet CartoDB Voyager Map Tiles (Live), HTML5 QR Camera Decoder (Live), GTFS Bus Position WebSocket Stream (Simulated).

• **Note on Configured / Planned Tech:** PostgreSQL/PostGIS, Redis, and physical NVIDIA Jetson hardware are configured options or planned future scope, while spatial queries currently execute via Python-side Haversine distance functions.

## 6. SIH Grand Finale Presentation Summary

**Problem:** Unverified civic officials exploit citizens, while municipal transport authorities lack real-time visibility into road hazards.

**Solution:** CivicShield provides a unified platform combining cryptographic QR identity verification with AI-powered mobile bus fleet urban sensing.

**Key Differentiator:** Converts existing public bus fleets into continuous mobile sensors with Edge AI processing, reducing cellular bandwidth usage by 99.98%.

## 7. Technical Accuracy & Factual Statement

**Claims Not To Make Yet:** Production municipal deployment, physical RTSP camera hardware streams, active PostgreSQL/PostGIS server deployment, or live Uber/Ola API keys.

**Strongest Factual Description:** *'The current CivicShield prototype is a functional, full-stack software application featuring a Next.js 14 frontend and a FastAPI (Python) backend. The platform provides working cryptographic QR identity verification against a database of government officials, weighted citizen trust scores filtered through an IsolationForest machine learning model to block rating manipulation, a Haversine-calculated government office locator, and a Gemini 1.5 Flash powered multimodal transit planner. For SIH PS 26124, it incorporates a dedicated Mobile Urban Sensing Dashboard demonstrating 4-camera bus telemetry, simulated Edge AI road hazard bounding box detection, and real-time Leaflet GIS mapping with automated municipal work order generation.'*